OPEN ACCESS
International Journal of High Speed Electronics and Systems
Vol. 35, No. 4 (2026) 2540536 (31 pages)
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DOI: 10.1142/S0129156425405364

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Artificial Intelligence-Driven Academic Performance Early Warning System for Engineering Universities: Application of XGBoost-SHAP Algorithm

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Received 25 February 2025
Revised 20 March 2025
Accepted 27 March 2025
Published 24 April 2025

Prior research concerning academic warning and performance prediction for university students within the contemporary credit system primarily utilizes statistical or machine learning techniques. Nevertheless, prediction models frequently encounter issues related to the excessive incorporation of various attributes, which culminates in intricate models and challenges in acquiring data time series. This investigation leverages students' historical academic records, employing machine learning interpretable methods to construct a prediction model. In dataset processing, an innovative application of hierarchical clustering was implemented, In this study, the HAC algorithm is compared with the DBSCAN algorithm, and the Silhouette score of the HAC algorithm is improved by 0.261; Calinski–Harabasz score increased by 154.615; Davies–Bouldin score decreased by 0.308, showing good model superiority. Therefore, students' actual academic conditions are considered to categorize regular graduating students. Subsequently, a feature fusion method was applied to extract dataset features. Ultimately, the XGBoost model was trained and evaluated, and the SHAP interpretable model was employed for the explanatory analysis of the developed model. Experimental outcomes demonstrate that, in comparison to Random Forest, Lasso, SVR, LightGBM, and Catboost machine learning algorithm models, the accuracy of XGboost prediction model is improved by 8.4%, 22.6%, 20%, 1.5%, and 5.9%, respectively. This underscores its efficacy in predicting a student's risk of extending their studies. The merits of this approach not only enhance the accuracy and

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interpretability of academic warnings but also propose a novel paradigm and trajectory for devising talent training programs in new credit-based engineering universities.

Keywords: Credit system; early warning evaluation; early warning of academic performance; SHAP; XGBoost model.

1. Introduction

1.1. *Background*

The credit system, originating in 18th-century Germany and further developed at Harvard University in 1872, aimed to address university survival and development issues. Initially implemented to resolve these challenges, it has since been adopted by numerous institutions globally due to its advantages in talent training, curriculum structure, and management mechanisms. Currently, the system's core elements include credit measurement units, an elective-based system enhanced by a tutorial approach, a central talent training framework, stringent management of education and teaching processes, and optimal utilization of teaching resources, finances, and manpower. However, the high implementation demands have led to challenges in some universities, including limited teaching resources, an incomplete academic guidance system, low student motivation, and inadequate management structures. Many scholars assert that families, schools, and society emphasize students' academic achievements and talent training quality. With the annual increase in higher education admissions, academic issues have emerged as a significant concern in higher education management, necessitating long-term strategies to address academic crises. These crises, becoming more severe, often trigger psychological and employment-related issues [1]. The evolving methods and means of student cultivation in higher education and changing learning styles complicate academic crisis management. Initially, students often struggle to adapt to new learning patterns, lack a comprehensive understanding of knowledge coherence and course systematicity, and exhibit cramming behavior before exams. Additionally, outdated professional knowledge, fragmentation, disconnects with emerging disciplines, and traditional teaching methods failing to engage modern students contribute to academic problems and crises. Furthermore, scholarly attention and theoretical research on academic crises remain insufficient. While some university administrators have initiated early warning research, these efforts lack a theoretical foundation and comprehensive approach to academic performance early warning [2]. The advent of the data era offers new approaches for enhancing student academic cultivation. Thus, research on early warning of academic performance is pivotal in improving talent cultivation quality in higher education institutions. Recently, machine learning has gained significant traction in various fields, including education, where it has been widely embraced. Machine learning technology can meet the needs of schools by analyzing and providing early warnings for students' academic performance, helping to identify students at risk of failure or dropout, and highlighting crucial factors affecting their performance. Constructing machine learning-based early warning

models for students facing academic difficulties is essential. This approach not only supports education managers in their roles but also significantly enhances teaching quality and learning outcomes.

1.2. *Related work*

Research on early warning of academic performance in higher education predominantly emphasizes comprehensive analysis of student performance data to identify correlations. Predominantly, predictions revolve around grade forecasting. Presently, four principal methodologies are employed to predict student performance: First, conventional methods predominantly use qualitative assessments through questionnaires, interviews, and surveys, lacking a foundation in analytical models and often yielding unidimensional conclusions. Second, statistical methods are frequently utilized, typically employing charts and summary data to depict data characteristics, encompassing frequency distribution analysis, dispersion analysis, and data distribution patterns. Descriptive statistical analysis represents the preliminary phase in data analysis, enabling an understanding of the fundamental characteristics of the data, which serves as a basis for more intricate subsequent data analysis. Third, data mining models, such as linear regression [3], decision trees [4], support vector machines [5], and neural networks [6], transform prediction problems into regression or classification problems. Fourth, grade prediction is treated as a user evaluation problem, employing recommendation-type techniques such as collaborative filtering and matrix factorization. Matthes and Stoeger [7] investigated the relationship between parents' learning abilities and children's academic performance, predicting children's academic performance based on parents' learning behaviors. The findings indicated that reducing parents' non-learning behaviors enhances children's academic performance. Gao [8] improved the Apriori algorithm to explore correlations within student behavioral data. Xuan *et al.* [9] utilized comprehensive data from 253 undergraduates to predict student performance, comparing six different base classifiers: random forest (RF), logistic regression (LR), support vector machines (SVM), Naive Bayes (NB), k-nearest neighbors (k-NN), and an ensemble learning method called stacking classifier. The findings revealed that the stacking classifier accurately predicted students' grade ranges. Shahzad *et al.* [10] conducted research on students' academic performance and mental health using artificial intelligence algorithms and social media. Feng *et al.* [11] analyzed learning behavior patterns using public education datasets from an educational data mining (EDM) perspective. They trained labeled data with six classification algorithms: J48, k-NN, Bayesian Networks, RF, SVM, and logit boost, establishing prediction models. The results showed that ensemble algorithms outperformed single classifiers, with logit boost exhibiting the shortest runtime among ensemble algorithms. Bressane *et al.* [12] employed artificial neural networks (ANNs) to identify association patterns between learning strategies, learning disabilities, and academic performance. Using cognitive characteristics, a fuzzy-based

AI model was developed. The research highlighted the significance of learning strategies in mitigating the adverse effects of learning disabilities on academic performance. Ou *et al.* [13] created a virtual educational robot system (AR-Bot) using augmented reality technology. They compared an experimental group of 41 first-year university students using AR-Bot with a control group of 34 first-year students using Scratch. The impact of both computational thinking (CT) tools on students' internal learning processes (learning enjoyment), CT skills (problem decomposition, algorithm design, and algorithm efficiency skills), and academic performance was assessed. The results showed that students using AR-Bot had higher learning enjoyment, algorithm design skills, and algorithm efficiency skills compared to those using Scratch, but not higher problem decomposition skills or academic performance. Matzavela [14] designed adaptive dynamic tests for assessing student learning outcomes, continuously comparing assessment results with decision tree algorithm results, and formulating prediction models for students' knowledge levels based on decision tree weights. Dol *et al.* [15] utilized EDM to classify, analyze, and predict students' academic performance, dropout rates, and teacher performance to enhance the teaching process. They also discussed future directions for EDM researchers, emphasizing the construction of models on datasets from educational environments to predict student performance for improving the teaching process. Ashraf [16] developed an accurate predictive teaching model using boosting techniques. Basic classifiers, including RFs, J48, k-NN, and NBs, were evaluated on a 10-fold cross-validation system. Both ensemble and filtering methods demonstrated significant improvements in predicting student performance compared to traditional classifiers. Additionally, after performance analysis of each method, two new prediction models were proposed based on improved results. Maniyan *et al.* [17] proposed a decision support system utilizing a hybrid data mining model to analyze educational data and uncover hidden patterns and rules. This system effectively mitigates or prevents issues such as dropouts and poor academic performance. Santos *et al.* [18] used one year of data from an information management school to build predictive models for two binary classification problems: identifying at-risk students and high-performing students. Traditional machine learning classifiers and long short-term memory units were tested at various stages of course completion. The best results were achieved using all course data, with the area under the curves of 0.756 and 78.2% for high-risk students using RF and extremely randomized trees, respectively. Early predictions could be made as early as 25% course completion. Buckley *et al.* [19] applied group-based trajectory modeling (GBTM) to learning management system data in online courses to determine predictors of successful course completion. The findings suggest that applying GBTM to LMS data, including synchronous participation information, can provide data enabling educators to identify early warning signs of potential student failure and intervene for those at risk. Malik *et al.* [20] proposed a machine learning technique to predict students' adaptation levels in online entrepreneurship education. Various algorithms, such

as RF, C5.0, classification, regression trees, and ANNs, were examined using the Kaggle Educational dataset. These algorithms recorded high accuracy rates and affirmed the ability of machine learning techniques to predict student adaptation to online entrepreneurship training. However, the analysis revealed that ensemble algorithms are seldom used. More traditional statistical analysis methods or machine learning methods like LR, SVM, and decision trees are often employed to warn students about their academic performance. These methods fail to consider dependencies between student characteristics or specifically analyze their impact on academic outcomes. Additionally, current algorithmic models lack interpretability, forcing university administrators to rely on experience for judgment, hindering the identification of the primary reasons for students' academic crises, and impeding the formulation of precise, timely intervention measures.

1.3. *Research motivation*

To address the aforementioned issues, ensemble learning algorithms have been extensively utilized. These methodologies achieve learning objectives through the construction of multiple learners, composed of a variety of base learners or individual learning models, including decision trees and neural networks. Ensemble learning leverages model fusion techniques to amalgamate the strengths of various models to produce the final classification decision. Recognized across numerous fields, classifier ensembles enhance robustness, stability, and accuracy by integrating multiple classifiers [21, 22]. The comparison of different machine learning algorithms is shown in Table 1. This research primarily elucidates the theories related to ensemble learning algorithms in academic early warning research, specifically focusing on the RF [23], XGBoost [24], LightGBM [25], and Catboost [26] algorithms, each offering unique advantages. The RF algorithm, part of the Bagging ensemble learning category, allows for the generation of multiple learners in parallel without interdependence among individual learners, thereby effectively handling high-dimensional data and exhibiting notable noise resistance. Conversely, the Boosting class algorithms, such as XGBoost, LightGBM, and Catboost, generate stronger learners through the sequential combination of weak learners, with individual learners exhibiting significant interdependence. XGBoost is distinguished by its predictive accuracy, efficiency, flexibility, and portability. LightGBM is noted for its accuracy and generalization capabilities, while Catboost excels in robustness, automatically managing missing or abnormal data, thus preventing overfitting during training. These four machine learning algorithms are characterized by robust learning capabilities and effectively resolve classification problems. Research by Nachouki *et al.* [27] introduced a model based on the RF method, identifying seven predictors from the academic records of 650 undergraduate computer science students. Their findings highlighted grade point average and high school grades as the primary predictors of course performance, with course category and attendance rate being equally significant. Miguéis *et al.* [28] analyzed first-year enrollment data to predict overall

academic performance, correlating academic success with average grades and time to degree completion. Their comparative study of SCMs, NB, decision trees, RF, boosting functions, and adaptive boosting decision tree algorithms revealed superior performance by RF and adaptive boosting decision trees. Bujang *et al.* [29] employed supervised machine learning methods to develop academic early warning models, demonstrating that the decision tree method achieved the highest accuracy (99.6%), effectively identifying students' academic risks. Okoye *et al.* [30] utilized advanced AI methods to predict student retention and graduation rates in higher education, analyzing retention ($n = 52262$) and graduation ($n = 53639$) data. Their machine learning model for retention and graduation data mining, using the Bagging method and 10-fold cross-validation, proved efficient and effective, achieving high precision (retention rate 0.909, graduation rate 0.822) and accuracy (retention rate 0.909, graduation rate 0.817). Chen *et al.* [23] proposed a novel approach for predicting academic performance by combining information gain (IG) and Laplacian score (LS) for feature selection. This methodology ranked features using IG and LS, followed by a sequential forward selection mechanism to identify the most pertinent indicators, and integrated RF with genetic algorithms for multi-class classification. Their case study demonstrated an average accuracy of 93.11%, improving baseline methods by at least 2.25%, and validated the approach's effectiveness through various evaluation metrics (accuracy, precision, recall, F-measure). Despite the widespread application and performance of these algorithms, challenges remain with high-dimensional categorical features affecting efficiency and performance, and the lack of consideration for dependencies between student characteristics in the aforementioned studies.

1.4. *Contributions*

The characteristics of current college student data, often encompassing a significant number of nominal categorical features and high dimensionality, are effectively addressed by the XGBoost algorithm. Renowned for its robust data fitting capabilities, XGBoost offers numerous parameters for fine-tuning model performance and provides benefits such as high accuracy, flexibility, and support for parallelization. Nevertheless, the absence of interpretability analysis in academic performance early warning models hinders educators' ability to pinpoint the primary causes of students' academic crises. As a result, precise and timely interventions cannot be formulated, and reliance on experiential summaries and recommendations persists. The SHapley Additive exPlanations (SHAP) interpretability method, introduced in this study, can mitigate the issue of poor interpretability in black-box models, thereby facilitating greater trust from decision-makers. SHAP has emerged as a research hotspot in academia in recent years due to its capability to enhance the interpretability of early warning models for academic performance. This method allows for an in-depth analysis of various influencing factors for students, examines the impact of dependencies among these factors on early warning outcomes,

Table 1. Introduction to Machine Learning Models.

Models	Description
XGBoost	XGBoost utilizes a gradient boosting framework to achieve high prediction accuracy. It combines the advantages of boosting algorithms and decision trees to optimize the objective function and handle complex feature interactions.
LightGBM	LightGBM is a high-performance gradient boosting framework that focuses on efficiency and accuracy in handling large-scale datasets. It utilizes a unique tree-based learning algorithm and offers superior speed and performance for solving machine learning tasks.
SVR	SVR is a machine learning algorithm that aims to predict continuous numeric values. It utilizes SVM to find a hyperplane that maximizes the margin around the predicted values, while allowing a tolerance for errors.
RF	RF model is an ensemble learning method that combines multiple decision trees to make predictions. It uses a combination of randomly selected features and bagging techniques to improve prediction accuracy and reduce overfitting.
Lasso	The Lasso model is a machine learning algorithm that performs both feature selection and regularization by adding a penalty term to the linear regression objective function. It encourages sparse solutions by promoting coefficients to be exactly zero, effectively selecting the most relevant features for prediction.
Catboost	Catboost is a gradient boosting algorithm that excels in handling categorical data. It provides state-of-the-art results by utilizing novel strategies such as ordered boosting, which optimizes the learning process. CatBoost is known for its efficiency, accuracy, and ability to handle complex datasets with categorical features seamlessly.

and mines crucial information from student data comprehensively. Consequently, it assists students in better self-management and enables educational administrators to understand students more effectively, thereby offering efficient guidance and ultimately improving the academic completion rates of college students. However, when applied to interpretable early warning models for academic crises in higher education, SHAP may face challenges from the extensive categorical data in campus big data, potentially affecting the model's accuracy and generalization ability. XGBoost, being a gradient boosting method, is adept at managing categorical features in this context. The primary contribution of this research is attributed to the integration of the XGBoost algorithm's strengths in managing extensive nominal categorical features and high-dimensional data within college student datasets, with the proficiency of SHAP interpretability methods in thoroughly examining the influencing factors and dependencies of student academic performance. An XGBoost-SHAP academic early warning method has been formulated to tackle the challenges posed by numerous categorical feature data in early warning of academic crises for students in newly established credit-based engineering universities. This approach not only offers an interpretable solution for academic performance early warning

in credit system-based colleges but also provides robust support for theoretically grounded academic research.

The structure of this paper is as follows. Sections 2 provides an introduction to the research methods; Sec. 3 presents the XGBoost-SHAP theoretical model for student academic prediction; Sec. 4 details the experiments conducted to validate the model's effectiveness, followed by the conclusions.

2. Research Methodology

The utilization of data mining driven by big data has become paramount in contemporary research. This exploratory analytical framework aims to quantitatively delineate and encapsulate intricate datasets, proving particularly efficacious when handling high-dimensional data replete with numerous nominal categorical features pertaining to college students. Free from the constraints of traditional research hypotheses, this approach significantly aids in the data preprocessing phase and facilitates the unveiling of intricate relationships among variables. Hierarchical clustering, a widely used unsupervised machine learning technique in cluster analysis, yields more intuitive and meaningful outcomes due to its hierarchical structure, which elucidates the hierarchical relationships within the data [29]. These hierarchical relationships reflect the associations and similarities between data points, allowing for a more profound comprehension of the data's internal architecture and distribution. The XGBoost algorithm is distinguished by its superior data fitting capabilities and flexible parameter control options, especially in managing categorical features. The SHAP interpretability method addresses the interpretability challenges of black-box models and has emerged as a prominent research focus in academia in recent years. The integration of XGBoost and SHAP algorithms enhances the interpretability of academic early warning models, enabling a comprehensive analysis of various factors influencing student academic performance and their interrelations. This method can effectively address the early warning of academic crises in categorical feature data, offering a novel paradigm and pathway for the development of talent cultivation programs in new credit-based engineering universities.

2.1. *Introduction to hierarchical clustering algorithm*

Hierarchical clustering algorithms, as opposed to traditional partition-based clustering methods, are predicated on the core principle of establishing hierarchical relationships among data points to execute clustering [31]. The outcome manifests as a tree-like structure, termed a clustering tree. This clustering tree distinctly delineates the clustering process and the interrelationships between the results, obviating the need to predetermine an often challenging number of clusters. Instead, it yields clustering results at varying levels of granularity. While other algorithms prioritize either the efficient completion of clustering computations or achieving results

akin to true classifications, they frequently encounter issues such as weak structural integrity in clustering outcomes, a lack of discernible representative points, and fixed granularity in cluster divisions. These limitations render clustering results less comprehensible for data analysts. When mining and analyzing extensive datasets, the structural attributes of the clustering process and outcomes, as well as the interpretability of clustering results, are essential for unearthing significant information and addressing fundamental data mining tasks. Consequently, hierarchical clustering algorithms, capitalizing on these advantages, have garnered particular favor in data analysis and mining, establishing themselves as indispensable tools in scientific research.

The fundamental concept of hierarchical aggregation clustering (HAC) involves iteratively merging the two nearest clusters to develop a hierarchical clustering tree structure. This study employs the HAC algorithm, encompassing the following steps:

(1) Initialization: Each feature vector is regarded as an individual cluster;
(2) The distances between all clusters are computed using the inter-cluster distance formula;
(3) The two nearest clusters are amalgamated;
(4) Step (2) is repeated until only a single cluster remains.

This methodology merges sub-clusters into a tree-like configuration, ultimately producing a clustering tree that includes all samples, commonly known as a dendrogram. The clustering tree does not directly yield clustering partition results. Typically, the clustering tree is segmented based on the dataset's characteristics and a predefined number of clusters, or the clustering process is halted early. The most prevalent approach is to partition the clustering tree by comparing inter-cluster distances or to prematurely cease the clustering procedure.

2.2. *Introduction to XGBoost algorithm*

XGBoost represents a type of boosting algorithm, fundamentally categorized as a boosting learning algorithm. Its underlying principle closely parallels that of gradient boosting decision trees (GBDTs). This algorithm enhances the GBDT algorithm by incorporating second-order Taylor expansion and regularization terms, effectively mitigating overfitting and expediting convergence. When compared to other algorithms, XGBoost exhibits advantages such as rapid execution, high efficiency, robust generalization capability, and suitability for handling large-scale tabular data. As a result, it has found extensive application in statistics and machine learning. The ensemble concept of the XGBoost algorithm involves constructing multiple weak estimators through the residual fitting, ultimately aggregating these weak estimators to form a strong estimator. With each inclusion of a weak estimator, the model's performance is anticipated to improve, enabling the final prediction

results to approximate true values as closely as possible while preserving maximal generalization capacity for unknown samples. The XGBoost algorithm can be expressed in an additive form as

$$\hat{yi} = \sum_{k=1}^{K} fk(xi), \quad fk \in F, \tag{1}$$

where $\hat{yi}$ represents the predicted value of the model; K represents the number of decision trees; f_k represents the k_{th} sub-model; x_i represents the i_{th} input sample; and F represents the entire set of decision trees. The objective function of XGBoost consists of two components: the loss function and the regularization term.

$$\mathrm{L}(\varphi)^t = \sum_{i=1}^{n} l(yi, \hat{yi}^{(t-1)} + ft(xi)) + \Omega(fk), \tag{2}$$

$$\Omega(fk) = \gamma T + \frac{1}{2}\lambda\|\omega\|^2, \tag{3}$$

where $\mathrm{L}(\varphi)^t$ represents the objective function for the tth iteration; $\hat{yi}^{(t-1)}$ represents the predicted values from the previous $t-1$ iterations; $\Omega(fk)$ represents the regularization term for the tth iteration model, used to mitigate overfitting; γ and λ represent the regularization coefficient to prevent the decision tree from becoming overly complex; and T represents the number of leaf nodes in the model. To further optimize the objective function, Taylor series expansion is applied to Formula (2), resulting in

$$\begin{aligned}\mathrm{L}(\varphi)^t &\cong \sum_{i=1}^{n}\left[g_i f_t(x_i) + \frac{1}{2}h_i f_t^2(x_i)\right] + \gamma T + \frac{1}{2}\lambda\sum\omega_j^2 \\ &\cong \sum_{j=1}^{T}\left[\left(\sum_{i\in I_j} g_i\right) + \frac{1}{2}\left(\sum_{i\in I_j} h_i + \lambda\right)\omega_j^2\right] + \gamma T,\end{aligned} \tag{4}$$

where g_i represents the first-order derivative of sample x_i; h_i represents the second-order derivative of sample x_i; ω_j represents the output value of the jth leaf node, and I_j represents the sample subset of the jth leaf node value. From this formula, it can be observed that the objective function is a convex function. By taking the derivative with respect to ω_j and setting the derivative equal to zero, the value of ω_j that minimizes the objective function can be obtained as follows:

$$\omega_j^* = -\frac{\sum_{i\in I_j} gi}{\sum_{i\in I_j} hi + \lambda}, \tag{5}$$

$$\mathrm{L}^{\sim}(\varphi)_{\min} = -\frac{1}{2}\sum_{j=1}^{T}\frac{(\sum_{i\in I_j} g)^2}{\sum_{i\in I_j} hi + \lambda} + \gamma T. \tag{6}$$

Formula (6) can be used to evaluate the quality of tree models, with lower values indicating better models. From this, the scoring formula (7) for node splitting in

trees can be derived. Finally, the Gain values of each feature are ranked to identify the optimal feature and the optimal split point.

$$\text{Gain} = \frac{1}{2}\left[\frac{(\sum_{i\in I_L} gi)^2}{\sum_{i\in I_L} hi+\lambda} + \frac{(\sum_{i\in I_R} gi)^2}{\sum_{i\in I_R} hi+\lambda} - \frac{(\sum_{i\in I} gi)^2}{\sum_{i\in I} hi+\lambda}\right] - \lambda. \tag{7}$$

The XGBoost model, distinguished from other models, employs a tree structure that segments variables into distinct regions, thereby maximizing the utilization of combinatorial variable effects to establish a more rational and accurate model structure. In the process of constructing decision trees, a penalty term for the tree structure is introduced. This term, adjustable through parameter tuning, partially mitigates multicollinearity and data imbalance issues, thereby enhancing model stability. The objective function is subjected to a second-order Taylor expansion, employing the second-order derivative in decision tree construction, which theoretically permits the extraction of more comprehensive information, thus improving classification accuracy. These attributes augment the algorithm's adaptability and practicality.

2.3. *Introduction to the SHAP interpretability method*

In the realm of machine learning and data science, model interpretability has persistently attracted attention from researchers and practitioners. The extensive application of intricate models, such as deep learning and ensemble methods, has amplified the necessity of understanding models' decision-making processes. SHAP, a Python library, is explicitly designed to elucidate the outputs of machine learning models. Rooted in the Shapley value theory from game theory, it achieves interpretability by analyzing SHAP values derived from the model. A primary merit of SHAP values is their precise reflection of each feature's influence on model predictions for individual samples, while effectively distinguishing between positive and negative impacts. By interpreting model predictions as linear functions of binary variables, a nuanced comprehension of each feature's contribution to model predictions is attainable. Employing SHAP necessitates initially selecting an appropriate explainer and training it with samples from the original model. This method not only addresses multicollinearity by assessing the impact of individual variables but also captures the synergistic effects among variables. Consequently, it comprehensively showcases the contributions of various parameter combinations to model predictions. This analytical proficiency furnishes significant guidance and strategies for model optimization.

The integrated tree model yields a probability value when engaged in classification tasks. SHAP has the capability to compute the Shapley value for each feature, quantifying its contribution to the ultimate prediction. Let g denote the explanation model, M the number of features, z the existence of a feature (taking values 0 or 1), and ϕ the Shapley value for each feature. The formula is presented as follows:

$$g(z) = \varphi_0 + \sum_{i=1}^{M} \varphi_i zi. \tag{8}$$

The SHAP value for each feature signifies the alteration in the model's prediction when that feature is conditioned upon. By evaluating these SHAP values, the contribution of each feature to the aggregate prediction can be ascertained, thus elucidating the differences between the model's predictions and actual outcomes. When $\varphi_i > 0$, it indicates that the feature exerts a positive influence on the predicted value; conversely, it implies a reduction in the feature's contribution. The strength of the SHAP model lies in its capacity to precisely mirror the influence of features in each sample and their respective positive or negative impacts on the prediction results.

This research develops a multi-feature fusion course association classification model and a student academic performance prediction model based on students' historical academic records. In the data processing phase, HAC is utilized to classify students who qualify for graduation, thereby establishing risk levels. The study examines course correlation analysis and applies the SHAP interpretable model for explanatory analysis. The detailed process of predicting student academic performance is depicted in Fig. 1. Initially, students' academic records are collected to construct the original dataset, followed by preprocessing, which mainly involves addressing missing values, outliers, and duplicate records, including imputing missing values, rectifying outliers, and eliminating invalid data. Subsequently, feature extraction is carried out to establish feature indicators for the dataset, and clustering methods are employed to group students' academic performance. Thereafter, machine learning models for early warning of academic performance are developed, with various machine learning algorithms compared and parameters optimized to identify the optimal regression model. The SHAP framework is then used to analyze the impact of feature attributes on early warning results within the model, offering three visualization management methods: overall interpretability analysis, variable interaction interpretability analysis, and local interpretability analysis. These methods provide a deeper understanding of the model's influence on academic prediction results and their features. Finally, the model is applied to predict and evaluate students' actual academic situations.

3. Validation of Interpretable Machine Learning Models

3.1. *Dataset development*

This research employed University C in Tianjin as the focal point. University C has been applying the credit system within its academic management framework since 2004. Over two decades of pedagogical practice, it has established an extensive array of management protocols and educational systems. In 2018, University C's H School was selected as part of the inaugural group of national-level "new engineering" initiatives in China. Consequently, the historical academic performance data

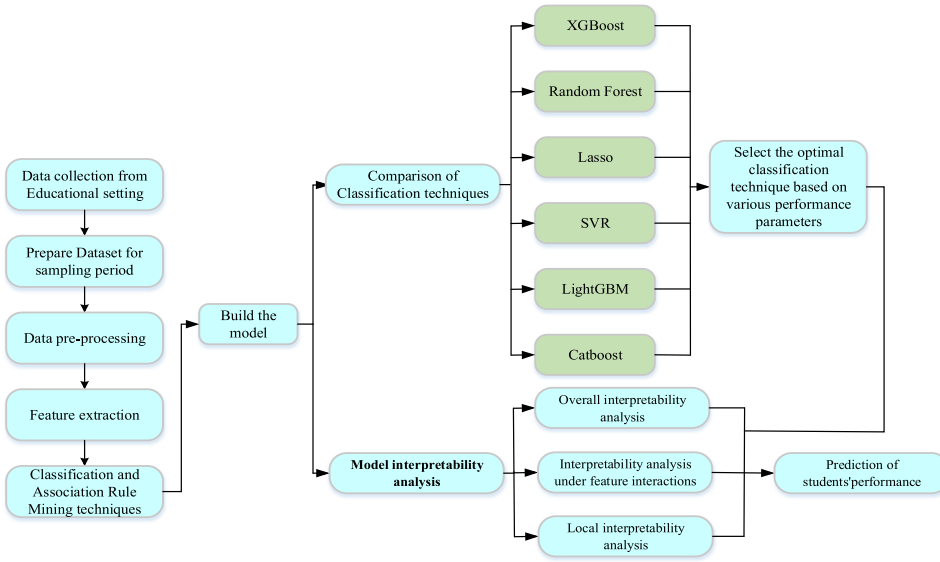


Fig. 1. Data mining process to predict students' performance.

of undergraduates from this school serves as vital data for pragmatic research on early warning evaluations and predictions of students' academic performance. In the realm of machine learning, data sources may encompass public datasets, experimental data, or questionnaire responses. When choosing data, it is imperative to evaluate the data's quality and suitability for the current task, as this directly influences the model's precision and effectiveness. This study constructed a dataset based on 1,313 undergraduate graduates from two cohorts, the Classes of 2018 and 2019, at School H of University C. Given the varied coursework undertaken by each student, suitable data processing tools were essential to standardize and harmonize the academic performance across all student samples, which included imputing missing values, addressing outliers, and handling duplicate records in the raw data. This entailed removing or correcting outliers and nullifying the impact of invalid data. To address privacy concerns, student names were anonymized using sequence numbers. Additionally, the dataset includes partial course grades for subjects such as Aircraft Reliability Engineering, Fundamentals of Aircraft Design, Theoretical Mechanics, Mechanics of Materials, and Mechanical Drawing. It also details whether students successfully completed their studies. Comprehensive information is provided in Table 2.

A direct evaluation method was used to assess the academic performance of various students to determine if the student could graduate on schedule. For those unable to graduate timely, their graduation status was denoted as 0. Those who could graduate timely were assigned appropriate grades. Additionally, to collect samples with similar traits, this study suggests employing hierarchical clustering

Table 2. Students' Partial Course Grades.

Name	Aircraft Reliability Engineering	Fundamentals of Aircraft Design	Theoretical Mechanics	Linear Algebra	Mechanical Drawing	Whether Successfully Graduated
1	72	65	72	60	72	Yes
2	76	60	62	82	70	Yes
3	73	61	64	72	71	Yes
4	60	68	64	73	69	No
5	85	60	60	65	76	No
6	71	71	87	65	92	Yes
7	64	72	80	60	84	Yes
8	65	60	75	70	82	Yes

Table 3. Classification of Student Academic Performance.

Scores	Grades
100	Excellent
80	Good
60	Pass
0	Fail

Table 4. Classification of Students' Risk Levels for Graduation.

Score range	Graduation Risk Level
0–60	Risk
60–80	General
80–100	Security

to categorize students who graduate on time. According to conventional grading practices, based on all academic achievements, students were classified into three categories: excellent, good, and pass. Scores were then allocated to these categories. Students who did not graduate on time were categorized as failures and recorded as 0. Table 3 presents the detailed outcomes.

To fulfill the requirement of predicting a student's ability to graduate within the allotted time, three risk categories (risk, general, and security) were delineated (Table 4). Consequently, in this research, the independent variables forming the dataset were derived through feature fusion and feature selection from the initial data. The dependent variables were established by clustering the academic performance of students who graduated on time into three levels and categorizing the performance of those who did not fail.

In the early warning evaluation of academic performance, the outcomes of HAC are more intuitive and easier to interpret, aiding in the identification of patterns and associations concealed within the data, thereby facilitating a better understanding

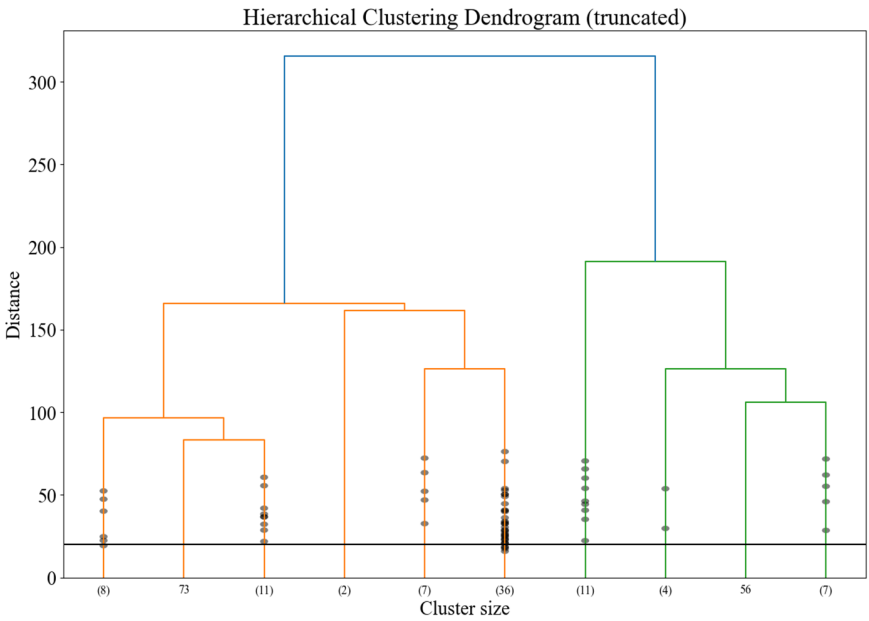


Fig. 2. Clustering results of academic performance classification for fresh graduates.

of students' learning situations and needs. HAC enables the segmentation of students into distinct groups or clusters based on their academic performance data. These clusters signify groups of students exhibiting similar academic performance. Through this methodology, groups of students potentially facing analogous academic challenges or displaying comparable performance can be discerned, enabling researchers to systematically comprehend the variances and commonalities among student groups, and thus develop targeted educational strategies and teaching plans. The tree diagram generated by HAC in this study is illustrated in Fig. 2. Three distinct clusters are clearly visible, with each cluster further subdivided into different categories (branches). The black circles on each branch indicate the number of samples, and each category can be considered to encompass samples with similar attributes, specifically students' academic performance. Consequently, samples within the same category can be randomly assigned. The labels derived through the clustering method can segment student academic grades more logically.

To validate the HAC algorithm's effectiveness, a comparison with the density-based spatial clustering of applications with noise (DBSCAN) algorithm is conducted. DBSCAN, a density-oriented spatial clustering technique, segments regions of adequate density into clusters and identifies clusters of arbitrary shapes within noisy spatial databases. A cluster is defined as the maximum set of density-connected points. As depicted in Fig. 3, the DBSCAN algorithm for student performance clustering results in merely two clusters, with black circles indicating invalid data. Moreover, the DBSCAN algorithm necessitates pre-setting two parameters:

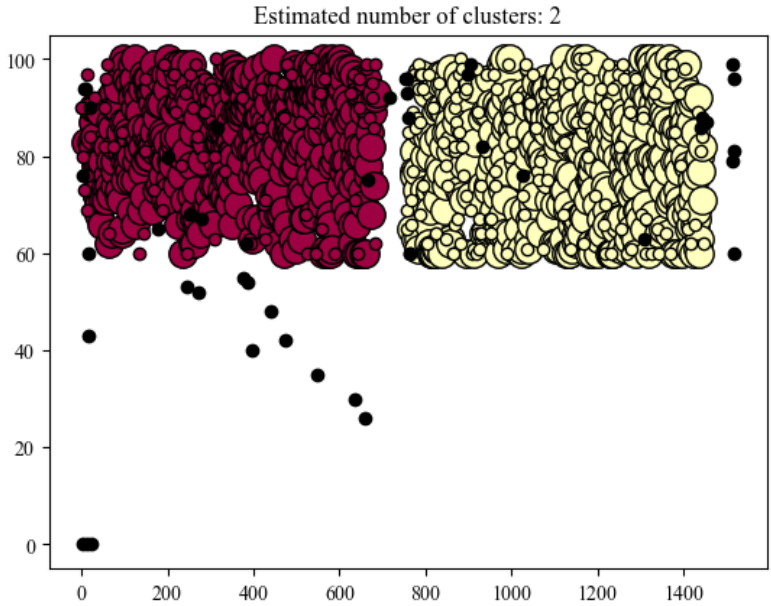


Fig. 3. Clustering results of DBSCAN algorithm.

neighborhood radius and minimum sample size, making it highly sensitive to data density and distribution, and it does not readily offer an intuitive display of clustering structures at varying levels. Conversely, the hierarchical clustering algorithm eliminates the need to pre-set the number of clusters, instead partitioning data through an inherent hierarchical structure. This results in a hierarchical clustering structure that vividly illustrates the inclusion and hierarchical relationships among different clusters, which is particularly advantageous for comprehending the similarities and differences between student groups. As shown in Table 4, a Silhouette score approaching 1 signifies better model performance. The Calinski–Harabasz score, derived from the silhouette coefficient and the sum of squared intervals, indicates superior model performance with higher values. A lower Davies–Bouldin score signifies enhanced model performance. The comparison in Table 5 suggests that, for complex academic early warning analyses, the DBSCAN algorithm is less suitable and interpretable compared to the hierarchical clustering algorithm.

Table 5. Comparison Results of Different Clustering Algorithms.

Algorithm / Parameter	DBSCAN Clustering Algorithm	Hierarchical Clustering Algorithm
Silhouette score	0.5344	0.795
Calinski–Harabasz score	1797.710	1952.325
Davies–Bouldin score	1.842	1.534

3.2. *Feature engineering*

In machine learning and data mining, feature selection refers to the process of identifying the most relevant and representative subset of features from the original data for model building or analysis tasks. This process directly impacts the model's performance and generalization ability. The main purpose of feature selection is to address the issue that original datasets may contain numerous features, not all of which significantly contribute to the target variable. By implementing feature selection, the number of features can be reduced, simplifying the problem's complexity while helping to improve the model's predictive ability, reducing the risk of overfitting, and enhancing predictive performance. In the context of predicting students' academic performance, this study primarily categorizes students' course grades into three types: individual course grades, category-based course grades, and elective course grades for different majors. During the feature selection process, it is essential to perform a specific fusion of category-based course grades and elective course grades for different majors. The specific methods are as follows:

(1) For courses within the same category (such as Further Mathematics A and B), the feature fusion method used is to take the average of all test scores in the same category, as depicted in formula (9).

$$w = \frac{m1 + m2 + \cdots + mn}{n}. \tag{9}$$

The average score w for courses of the same category is calculated, where m_n represents the score for each course within the same category, and n denotes the total number of courses in that category. It is assumed that student S's grades in Further Mathematics A and B are 95 points and 85 points, respectively, the further mathematics score obtained through feature fusion is 90 points.

(2) To maintain the dataset structure's consistency and accurately predict the courses' impact on early academic performance warnings and extended study duration, a series of fusion processes must be applied to the original data features. This is particularly critical due to the significant influence of elective courses on students' study extension and the varied nature of course selection based on individual interests. The weighted average can be calculated as the final input feature, as shown in Eq. (10):

$$V_i = \frac{\sum v_{i,j} x_j}{\sum x_j}, \tag{10}$$

where j represents major electives, x_j represents the credit of the course, and $v_{i,j}$ represents the examination score of a particular student in that course.

Feature attributes with extremely low failure rates, such as college physical education and ideological and political theory courses, were directly eliminated. Ultimately, 14 courses were identified as feature indicators through feature fusion methods, and a correlation analysis was conducted (Table 6).

Table 6. Feature Indicators of the Datasets.

Feature Indicators
C Language
Complex Functions
Probability Theory
Mechanical Control
Introduction to Civil Aviation
Mechanics
College English
Electrotechnics
Further Mathematics
Mechanical Drawing
General Physics
Physics Experiments
Linear Algebra
Major Electives

The Pearson correlation coefficient matrix is a statistical tool utilized to assess the linear correlation between variables. Each element in the matrix signifies the Pearson correlation coefficient between two variables. A coefficient of 1 denotes a perfect positive correlation, whereas a coefficient of -1 denotes a perfect negative correlation. A coefficient of 0 signifies no linear correlation. In this study, the Pearson correlation coefficient matrix was calculated to quantify the associations between features. The results are depicted in Fig. 4. Darker colors in the Pearson correlation coefficient matrix indicate higher correlations. Generally, when the coefficient exceeds 0.85, only a single feature with a higher correlation to the target variable needs to be retained. The feature correlation analysis revealed that the correlations among the features were not strong. Consequently, there was no necessity to eliminate any features.

3.3. *Development of machine learning models*

3.3.1. *Model parameter optimization*

In the investigation of student performance prediction, the grid search method is utilized in this study to optimize XGBoost algorithm parameters, with the goal of augmenting prediction accuracy and generalization capability. By employing a grid search for XGBoost parameter optimization, the parameter space is meticulously explored, enabling automated parameter tuning, precise evaluation of model performance, enhancement of prediction accuracy, and provision of flexibility. This methodology aids researchers in constructing and optimizing prediction models more effectively, thereby advancing the understanding and forecasting of trends and changes in student performance. The parameter optimization procedure for the XGBoost algorithm is depicted in Fig. 5. The horizontal axis denotes the number of training iterations, while the vertical axis represents the error value. As illustrated in the figure, the error value diminishes progressively with an increase in the

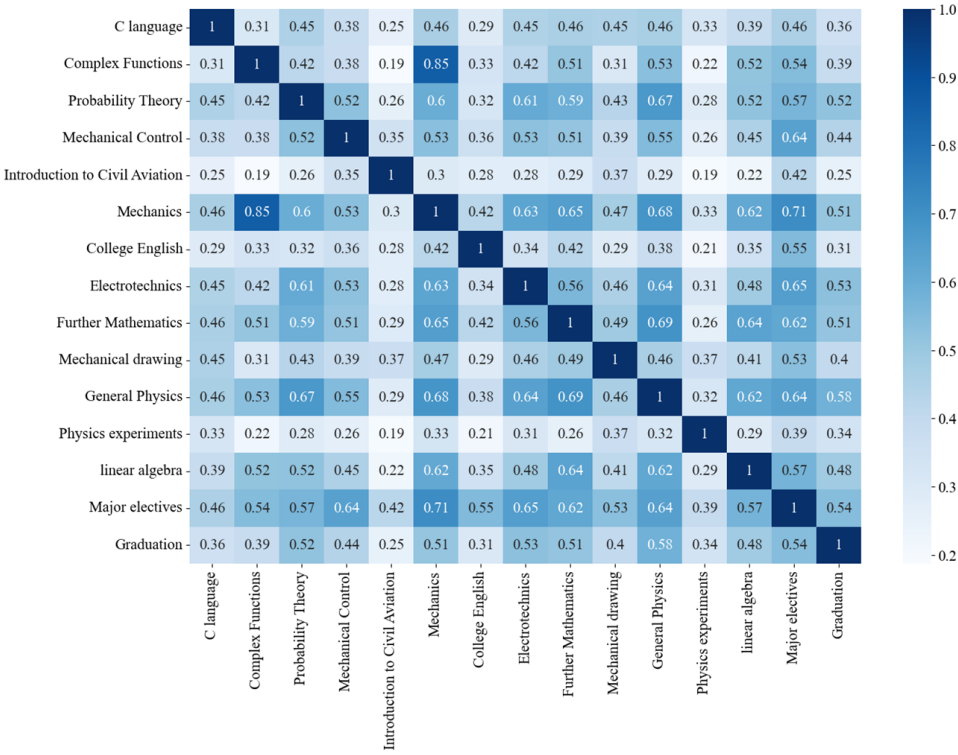


Fig. 4. Characterization of Pearson correlation coefficient.

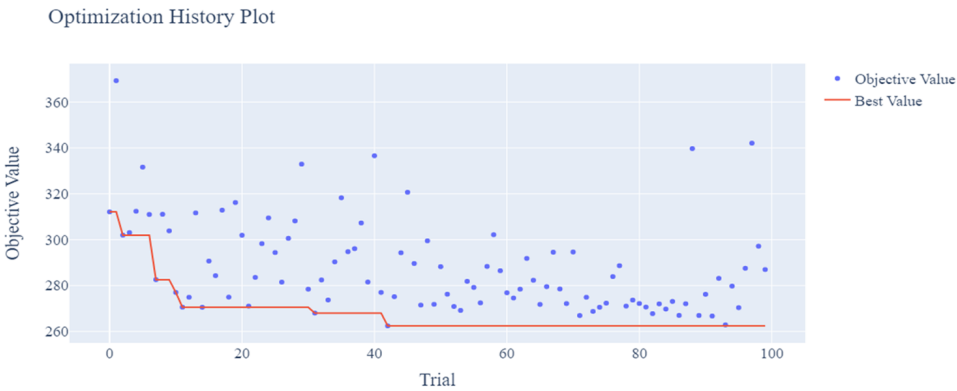


Fig. 5. Parameter optimization process of XGBoost algorithm.

number of training iterations. The solid red line in the graph signifies the optimal solution for the optimization results. Table 7 exhibits the final optimization outcomes for various XGBoost algorithm parameters, listing the specific optimal values for crucial parameters such as n_estimators, max_depth, and learning_rate.

Table 7. Parameter Optimization Results for XGBoost Algorithm.

Parameter	Investigated Parameters	Optimal Parameters
n_estimators	range (100, 1000)	952
max_depth	range (3, 10)	8
learning_rate	loguniform (1e-3, 1e-1)	0.0325
subsample	uniform (0.6, 1, 0.1)	0.6
colsample_bytree	uniform (0.6, 1.0)	1.0
gamma	loguniform (0.01, 1)	0.244
reg_alpha	loguniform (0.01, 1)	0.361
reg_lambda	loguniform (0.01, 1)	0.213
min_child_weight	loguniform (1, 10)	1
random_state		42
n_jobs		−1

3.3.2. *Comparison of different models*

In Python, the relevant libraries and datasets for model training and prediction were initially imported. A random selection process was employed to allocate 80% of the data from the training set and test set to the training set, with the remaining 20% designated for the test set, thereby maintaining a 4:1 ratio between the training and test sets. The performance of the predictive machine learning model was determined by evaluating the mean-square error (MSE) [32], root mean squared error (RMSE) [33], mean absolute error (MAE) [34], mean absolute percentage error (MAPE) [35], and the coefficient of determination (R^2) [36] on the test set. MSE, a commonly utilized metric, quantifies the difference between model predictions and actual observations, thereby assessing the model's fit on the given data. This metric is derived by averaging the squared differences between predicted and observed values. RMSE, another frequently employed indicator, assesses the discrepancy between model predictions and actual observations, providing a more intuitive representation of the magnitude of prediction errors by calculating the square root of the mean squared differences. MAE measures the absolute differences between predicted and observed values, offering a straightforward assessment of model fit. MAPE, calculated by converting the absolute differences between predicted and actual values into percentages, represents the average error magnitude and serves as a powerful indicator for enhancing data-driven decision-making accuracy. R^2, representing the proportion of variance in the dependent variable explained by the model, serves as an indicator of the goodness of fit for regression models, with higher R^2 values indicating better model fit. Generally, smaller RMSE, MSE, MAE, and MAPE values denote superior performance, while R^2 values closer to 1 indicate better results. This study utilized the XGBoost algorithm model for practical application. To compare the effectiveness of different machine learning models in academic performance prediction, simulations were conducted under various models, including RF, Lasso, SVR, LightGBM, and Catboost. The accuracy comparison for different models is depicted in Table 8. The figure illustrates that in comparison with other model parameters, XGBoost has the smallest error between predictive

Table 8. Comparison of Evaluation Metrics for Different Models.

Models	RMSE	MSE	MAE	MAPE	R^2
XGBoost	16.199	262.437	7.831	1.403	0.852
Random Forest	17.256	270.540	7.921	1.623	0.768
Lasso	18.547	268.420	8.450	1.525	0.626
SVR	16.725	265.428	8.236	1.479	0.652
LightGBM	17.865	272.430	7.987	1.584	0.837
Catboost	17.228	296.809	8.616	1.782	0.793

value and actual observations, the XGBoost model exhibits the best performance, with an R^2 of 0.852, achieving an accuracy of 85.2%. Hence, XGBoost is deemed the most effective machine learning model for this application.

3.4. *Model interpretability analysis*

Machine learning models are often perceived as "black box processes", in which the relationship between features and output variables remains implicit. SHAP interpretable models can be employed for visual analysis to explore the relationship between inputs and outputs in depth. SHAP is utilized to elucidate the predictions of machine learning models and can interpret the output of any machine learning model. By assigning importance scores to each feature for a specific prediction, SHAP facilitates a better explanation and comprehension of the model's behavior. SHAP values provide insights into how each feature contributes to the overall prediction, making it an invaluable tool for model interpretation and understanding [23]. This study implemented the SHAP framework to examine the impact of feature attributes on students' early academic warning results. Furthermore, three visualization management methods — overall interpretability analysis, variable interaction interpretability analysis, and local interpretability analysis — were provided. These methods enhance the understanding of the model's predictions and the influence of features on the prediction results.

3.4.1. *Overall interpretability analysis*

SHAP offers a comprehensive visualization technique that displays the SHAP values for all samples within a graph, with each feature's SHAP value represented by a color to indicate its relative significance. This method allows for a more profound comprehension of feature importance, feature interactions, and additional model information, thereby augmenting the interpretation of machine learning models. Figure 6(a) illustrates the mean SHAP values across all features for the entire dataset. The results indicate that courses such as Major Electives, General Physics, Further Mathematics, Electrotechnics, and Linear Algebra have a more significant effect on students' early academic warnings and study period extensions, whereas Mechanical Drawing and Introduction to Civil Aviation possess lower mean SHAP values, suggesting a lesser influence on study period extensions. Consequently, universities should prioritize students' education in major electives and foundational

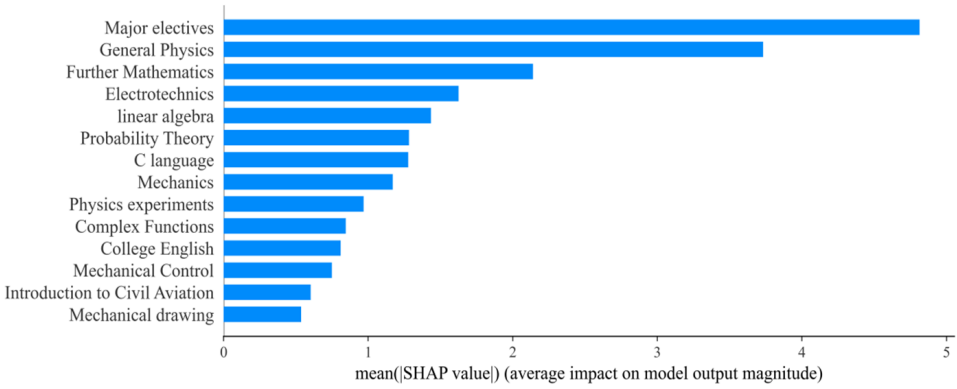
mathematics and science courses. While the ranking of feature importance reveals the significance of different features in the early warning results, it does not clarify whether a student's feature value exerts a positive or negative impact on their academic outcomes. Figure 6(b) presents the SHAP values for all samples across various features using a scatter plot, which displays the SHAP value of each feature for each sample, elucidating the positive or negative effects on students' early academic warning outcomes. Each dot represents a sample; regions with dense dots indicate a high concentration of samples. The X-axis denotes the SHAP value, reflecting the impact of student features on their academic success. The right side of the graph shows the magnitude of feature attribute values, with red indicating high eigenvalues and blue indicating low eigenvalues. The effect on the target variable can be inferred from the SHAP value magnitude. For instance, considering General Physics, a high eigenvalue corresponds to a SHAP value greater than 0, whereas a lower eigenvalue significantly enhances its positive impact on students' study period extensions. Specifically, students achieving higher scores in General Physics exhibit a reduced risk of extended study periods.

3.4.2. *Interpretability analysis under feature interactions*

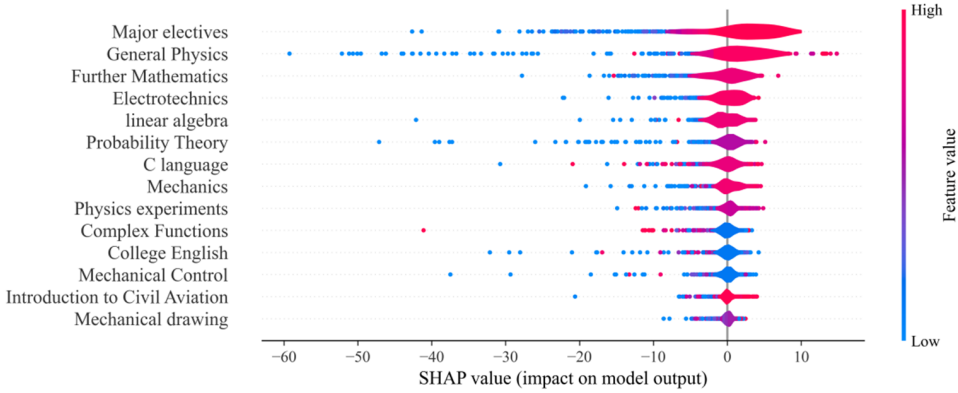
Interaction effects on the target variable can arise from multiple variables. SHAP interaction values measure the interaction effect between two features, highlighting their synergistic or antagonistic impact on the predicted outcome. Through the analysis of SHAP interaction values, hidden feature relationships within the model can be revealed, thereby enhancing its interpretability and credibility. Figure 7 demonstrates the relationship between the values of all features and the SHAP values for each feature across the entire sample.

The X-axis of the plot indicates the SHAP value for each feature, while the Y-axis represents the value of each feature. The color intensity reflects the correlation strength, with blue denoting a positive correlation and red indicating a negative correlation. Darker shades signify a higher degree of correlation. By scrutinizing the correlation coefficients, the influence of different features in determining the target variable can be discerned. Major Electives, Electrotechnics, Mechanics, Further Mathematics, and General Physics showed strong positive correlations with other courses. Conversely, Complex Functions, College English, and Introduction to Civil Aviation exhibited significant negative correlations with other courses.

The impact of individual features can be elucidated using SHAP visualizations with variable interactions, where scatter plots are utilized for representation. As illustrated in Fig. 7, a scatter plot showcasing the contribution of a single feature alongside another feature employed for coloring to the predicted value enables further exploration of interaction effects between individual features and their changing trends on the predicted value. The X-axis represents the eigenvalue, while the Y-axis signifies the corresponding SHAP value, with each point indicative of a sample. In the scatter plot, the color and size of points typically denote the eigenvalue



(a)



(b)

Fig. 6. Impact of features on model output. (a) Feature Importance Ranking based on SHAP. (b) Feature Impact based on SHAP.

of the data point and its frequency within the entire dataset. This approach aids in better comprehending the relationship between the feature and the target variable. Univariate SHAP visualizations provide an intuitive understanding of each feature's impact on the target variable, while also examining issues such as nonlinear relationships or outliers. Concurrently, in the SHAP plot of feature interactions, the value of a particular sample of the interacting element can be swiftly identified through the trend of the SHAP value of a specific feature. Compared to the univariate visualization of the target variable, this method more intuitively reflects the trend of feature impacts on students' graduation levels.

Utilizing Fig. 8(a) as an illustration, the impact of the interaction between the features of Major Electives and Further Mathematics on early academic warning outcomes was thoroughly analyzed. An elevated score in Major Electives corresponds to a larger SHAP value, signifying a greater impact on the student's

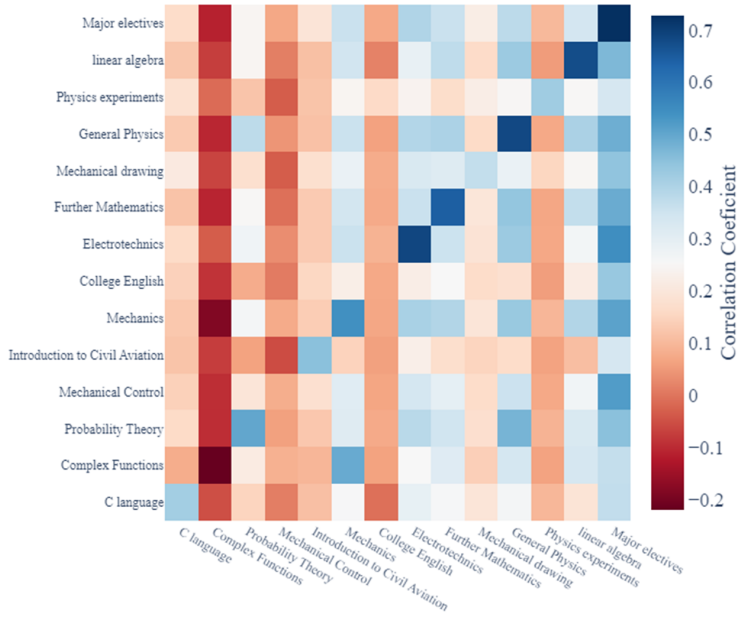


Fig. 7. Relationship between eigenvalue and SHAP values.

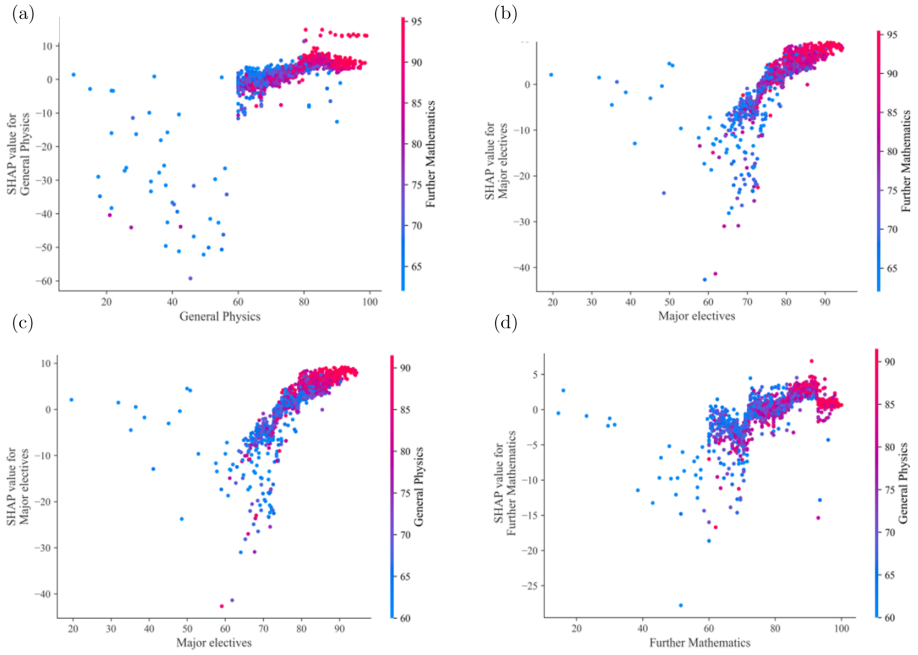


Fig. 8. SHAP feature dependence. (a) The influence of Further Mathematics on the characteristics of Major Electives. (b) The influence of Further Mathematics on the characteristics of General Physics. (c) The influence of General Physics on the characteristics of Major Electives. (d) The influence of General Physics on the characteristics of Further Mathematics.

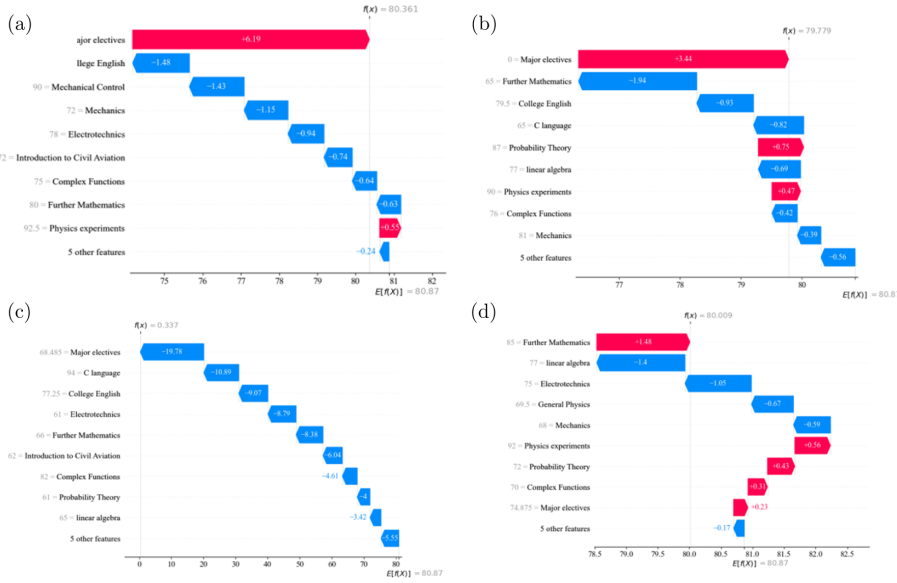


Fig. 9. Local interpretability of SHAP values for samples. (a) Local interpretability of SHAP values for sample A. (b) Local interpretability of SHAP values for sample B. (c) Local interpretability of SHAP values for sample C. (d) Local interpretability of SHAP values for sample D.

likelihood of successfully completing their studies. Additionally, performance in Further Mathematics is vital in determining timely graduation. Figure 8(a) serves as a SHAP feature dependence plot, primarily focusing on Major Electives scores, with red coloring employed to highlight potential interactions between the two features. When the eigenvalue of Further Mathematics surpasses 85, the red hue predominantly appears in the upper regions of Major Electives scores. By integrating this with early warning results, it is inferred that superior performance in Further Mathematics generally correlates with higher Major Electives scores, thereby increasing the probability of study completion. Analyzing variable interactions further elucidates the model's output, examining the influence and contribution of each input feature on the outcome. This process augments confidence and comprehension of the model, reveals its strengths and weaknesses, aids in model and feature refinement, and offers more coherent explanations to model users or audiences.

3.4.3. *Local interpretability analysis*

To gain a deeper comprehension of the predictive or decision-making processes of machine learning models for specific data points or groups of data points, explanations for local samples were provided. Particularly, for a given sample with parameters, the contribution to the target variable can be depicted by computing the SHAP value for each feature. Figure 9 illustrates a local interpretability analysis showcasing SHAP values across various samples.

Table 9. Contribution Values of All Features to the Target Variable for Local Samples.

Parameter Value	Base Value	SHAP Value	Output Value
Major Electives = 84.14		+6.18	
College English = 84.5		−1.48	
Mechanical Control = 90		−1.43	
Mechanics = 72		−1.15	
Electrotechnics = 78	+80.87	−0.94	+80.36
Introduction to Civil Aviation = 72		−0.74	
Complex Functions = 75		−0.64	
Further Mathematics = 80		−0.63	
Physics Experiments = 92.5		+0.53	
Other Features		−0.24	

Table 10. Comparison of Sample Predicted Values in Fig. 9.

Sample	Predicted Value	True Value	Graduation Risk Level
1	80.36	80	Safety
2	79.78	80	General
3	0.337	0	Fail
4	80.01	80	Safety

Taking Fig. 9(a) as an illustration, the SHAP value for each feature of the sample was computed to elucidate the contribution of the current sample to the target variable. This approach facilitated a quantitative analysis of the sample, accurately predicting the likelihood of a student's early academic warning. The sample's average SHAP value was determined to be 80.87, with a predicted value of 80.36, signifying a positive impact on early academic warning and indicating that it will not cause an early academic warning. Furthermore, College English, Mechanical Control, Mechanics, Electrotechnics, Introduction to Civil Aviation, Complex Functions, and Further Mathematics were identified as having a negative impact, ranked in descending order of influence. Specifically, they negatively affect the early academic warning of the sample, whereas Major Electives and Physics Experiments exert a positive impact. Table 9 details the specific contribution values of all features of the local sample to the target variable, and Table 10 compares the predicted values of several samples to ascertain students' graduation risk levels. In practical applications, to predict a student's academic performance accurately based on given parameters, the output value is derived by summing the base value (the model's expected output value) of each independent variable with the SHAP value, thereby making a quantitative prediction of the student's academic performance and early warning.

4. Conclusion and Recommendations

4.1. *Conclusion*

The study explored the early warning prediction of academic performance for all graduates of the classes of 2018 and 2019 in School H of University C. The student academic performance dataset was segregated into grades and categorized by

graduation risk levels. The findings indicate that in comparison to Random Forest, Lasso, SVR, LightGBM, and Catboost machine learning algorithm models, the accuracy of the XGboost prediction model is improved by 8.4%, 22.6%, 20%, 1.5% and 5.9%, respectively. This underscores its efficacy in predicting a student's risk of extending their studies. This emphasizes that XGboost has the highest performance and can be regarded as the optimal machine learning model. The output from the machine learning model was visualized and analyzed using SHAP interpretability, and conclusions were derived accordingly.

Initially, comparing the test sets demonstrated that major electives, general physics, further mathematics, electrotechnics, and linear algebra significantly impact students' study period extension and have the greatest influence on the target. Courses such as Complex Functions, College English, and Introduction to Civil Aviation exhibit a lesser effect in aligning. Majors related to new engineering courses, such as Further Mathematics and Physics, impose higher foundational requirements on students. If students' academic foundations are weak, they are more likely to face the risk of extended study periods.

Subsequently, further analysis was conducted based on feature interactions and local interpretability. This investigated the influence of the interaction between two features on students' early academic warning results. When course grades are higher and SHAP values are larger, the impact on students' successful completion of their studies is more pronounced. Additionally, when two important features interact, the higher the grades, the stronger the positive effect on students' successful completion of their studies. Specifically, this is beneficial for students to graduate, as can be observed through the SHAP value plot.

Lastly, in practical applications, to predict students' academic performance accurately, the base value (the model's expected output value) for each independent variable was calculated using the provided parameters, combined with the SHAP value to equal the final output value. By combining the SHAP value of each feature, it becomes possible to quantify the impact of each independent variable on the final prediction result, thereby offering a more comprehensive understanding of the model's prediction mechanism.

4.2. *Research limitations and prospects*

The XGBoost-SHAP algorithm significantly enhances transparency and interpretability in predictive modeling, enabling a deeper understanding of the model's operational mechanics. Additionally, it aids in developing more precise academic intervention strategies, thereby providing novel perspectives and methodologies for future research. This method introduces an innovative paradigm for constructing early warning mechanisms and designing talent cultivation plans. Nevertheless, this study is limited to the application of the XGBoost algorithm in academic prediction research. Further research is required for the design and implementation of multi-algorithm integrated student academic assessment models and early warning

systems. Initially, the integration of various data sources is essential. Different data sources provide multidimensional insights into students' learning, behavior, and background. Typical data sources include academic records (course grades, credit completion status, academic warning history), learning behavior data (study time, on-campus learning activities, online learning platform engagement), social data (club participation, social media activity), and personal background information (family circumstances, pre-enrollment academic, and social background). The comprehensive use of these data sources can create a more complete and diverse student profile, aiding in more precise prediction of academic early warning scenarios. Furthermore, multi-algorithm fusion can combine the strengths of various algorithms, improving the robustness and accuracy of prediction models. Model stacking can also be utilized, where the predictions from multiple base models serve as new features for a secondary model to make the final prediction. Model stacking can better capture nonlinear relationships and complex patterns in the data, thereby enhancing prediction accuracy. Finally, continuous improvement and monitoring are critical. The establishment of an academic early warning system is a continuous enhancement process. By continuously collecting and analyzing student data, timely adjustments to models and algorithms can be made to address changes in student learning status and new situations. Regular updates to models and algorithms ensure the prediction system's effectiveness and practicality. In summary, utilizing multi-algorithm fusion and integrating different data sources are key strategies for enhancing the effectiveness of university student academic early warning systems. This approach allows for more accurate prediction of potential academic difficulties students may face, enabling timely intervention measures to help students improve their learning conditions.

Acknowledgment

This study was supported by the Fundamental Research Funds for the Central Universities Civil Aviation University of China Special No. 3122025015.

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